



Stroke

Stroke is an emergency condition that occurs when the blood flow to the brain is suddenly interrupted. The blood vessel is either blocked or ruptured. Brain cells cannot receive oxygen and nutrients and die within a short period of time. This leads to impairment of functions such as speech, movement, vision, or balance.

There are three main types of stroke: 1) Ischemic stroke: The most common type. A blood vessel in the brain becomes blocked by a clot or narrowing. 2) Hemorrhagic stroke: A blood vessel in the brain ruptures, causing bleeding. 3) Transient ischemic attack (TIA): Blood flow is interrupted for a short period of time. The symptoms usually disappear within a few minutes or hours and do not cause permanent damage. However, it may be a warning sign of a major stroke in the future.

In 2021, there were approximately 93.8 million people living with stroke worldwide, while 11.9 million new cases were recorded in the same year. Stroke is one of the leading causes of death and disability. Approximately one in four adults is at risk of having a stroke during their lifetime. This risk has increased by 50% over the past 20 years.

The burden of stroke, including years of life lost due to death and disability, has also increased. The majority of deaths and disabilities occur particularly in low- and middle-income countries. The global economic cost of stroke exceeds 890 billion dollars.

The symptoms of stroke usually occur suddenly. The most common symptoms include: drooping on one side of the face, weakness in an arm or leg, impaired speech or speech that is difficult to understand, sudden loss of balance or dizziness, and sudden loss of vision.

Symptoms vary depending on which part of the brain is affected. The right hemisphere controls the left side of the body, while the left hemisphere controls the right side. The location and extent of the damage determine how severely the person will be affected. Some strokes do not cause obvious symptoms; these are referred to as “silent strokes.” They may increase the risk of dementia.

The symptoms of TIA are short-lived and disappear completely. Nevertheless, it still requires emergency medical attention.

The most important modifiable risk factor for stroke is high blood pressure. People with high blood pressure have an almost threefold increased risk of stroke. Other important factors include: smoking, high cholesterol, diabetes, obesity and physical inactivity, a diet high in salt, harmful alcohol use, air pollution, and kidney problems.

Non-modifiable factors include age- the risk increases every ten years after the age of 55- having previously experienced a stroke, and certain heart conditions, such as atrial fibrillation.

Stroke is less common in young adults (18-55 years), but its causes are different. In this age group, arterial dissection, an opening in the heart (patent foramen ovale), certain coagulation disorders, or rare forms of vascular inflammation are more frequently involved. In older adults, high blood pressure, atherosclerosis, and heart-derived blood clots are more prominent causes.

Brain cells require continuous blood flow. When blood flow is interrupted, the cells die. In ischemic stroke, a blood vessel becomes blocked by a clot or plaque. The clot may form in the brain or may come from another part of the body. In hemorrhagic stroke, a weakened blood vessel ruptures; high blood pressure gradually weakens blood vessels over time.

Different regions of the brain control different functions. Depending on the area that is damaged, speech, movement, vision, or thinking may be affected.

If a stroke is suspected, the patient should be taken to the hospital immediately. Brain imaging, such as CT or MRI, is performed for diagnosis. This determines whether the cause is bleeding or blockage. In young patients, additional

investigations may be required to identify the underlying cause.

Time is particularly critical. In ischemic stroke, clot-dissolving medication, such as tPA, may be administered within the first few hours after the onset of symptoms. In some patients, a procedure to remove the clot from the blood vessel (thrombectomy) may be performed. In hemorrhagic stroke, blood pressure is controlled, and surgery may be considered if necessary.

Patients have better outcomes when they are monitored in specialized stroke units. Once the patient is stable, rehabilitation begins, including physical therapy, speech therapy, training in activities of daily living, and psychological support.

During the first few days after a stroke, complications such as brain edema, difficulty swallowing, pneumonia, seizures, or shoulder pain may occur. In the long term, permanent limitations in movement, speech disorders, memory problems, depression, and loss of independence may develop. Stroke increases the risk of dementia by more than twofold. Silent strokes may also contribute to dementia.

In young patients, standard approaches may not be sufficient to determine the cause of stroke; specialized investigations may be required. In addition, the relationship between stroke and dementia is becoming increasingly well understood. Controlling common risk factors- such as high blood pressure, high cholesterol, diabetes, and physical inactivity- can help reduce the risk of both stroke and dementia.

In conclusion, stroke is largely a preventable condition. Controlling high blood pressure, quitting smoking, maintaining a healthy diet, and staying physically active are the most effective steps. Recognizing the symptoms and seeking immediate medical assistance can increase the chances of survival and recovery.

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